

Automated Detection of Gen Z Slang in Social Media Texts

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Abstract

Slang detection in social media text is challenging due to rapidly evolving vocabulary and pervasive polysemy, where terms carry both literal and slang interpretations depending on context. We address Gen Z slang detection under two task formulations: sentence-level binary classification and token-level sequence labeling using BIO tags. We construct a balanced dataset of approximately 20,000 examples and evaluate four approaches: dictionary lookup, TF-IDF with logistic regression, fine-tuned BERTweet, and few-shot and chain-of-thought prompting with DeepSeek V3.2 and Qwen 2.5 Plus. Fine-tuned BERTweet achieves the strongest performance (sentence-level $F1 = 0.911$, token-level $F1 = 0.785$), and error analysis reveals that polysemous words remain difficult across all systems, highlighting the need for improved contextual sense disambiguation in informal language processing.

1 Introduction

Language on social media evolves rapidly, with new Gen Z slang terms emerging daily, making it difficult for both humans and NLP systems to keep up with their meaning and usage. Many of these terms exhibit polysemy, where a word has both a dictionary definition and a slang interpretation, such as “fire” referring to combustion or praise. This semantic ambiguity requires LLMs to rely on contextual understanding and knowledge of emerging vocabulary.

In this project, we address Gen Z slang detection as a binary classification and sequence labeling problem. By leveraging specific datasets, we developed models to identify if a sentence contains slang and locate the specific slang tokens. This aims to evaluate how well modern neural sequence models capture the evolution of informal language.

2 Related Work

Slang Detection and Identification Pei et al. (2019) established an early framework for slang detection using BiLSTM-MLP and BiLSTM-CRF models, demonstrating that CRF-based models perform best for sentence-level detection ($F1 \approx 0.8$) while BiLSTM-MLP with Char-CNN is stronger for token-level identification ($F1 \approx 0.5$). Their work frames slang detection as both classification and sequence labeling problem, which we adopt. However, these sequential models lack the contextual depth of modern transformers, and their datasets predate much of today’s Gen Z vocabulary that motivates our work.

Informal Language Processing in LLMs Sun et al. (2024) evaluated LLMs on the OpenSub-Slang dataset, finding strong zero-shot performance from GPT-4 ($F1 \approx 0.84$), and the best overall results from fine-tuned GPT-3.5 ($F1 \approx 0.90$). This suggests LLMs already possess substantial implicit slang knowledge, but still benefit from fine-tuning for token-level precision. We extend this work to contemporary internet slang from Gen Z and Gen Alpha communities, which differs substantially from the movie subtitle-derived slang in OpenSub-Slang. We also directly incorporate LLM prompting as one of our four evaluated approaches, allowing us to benchmark frontier models against supervised baselines on the same dataset.

Fine-tuning BERT for Slang Most closely related to our supervised approach is Mohammed and Shafqat (2024), who fine-tuned BERT for both binary detection and BIO token tagging. Despite a high weighted $F1$ of 0.97, their macro $F1$ of 0.82 and low I-token $F1$ of 0.54 reveal that the model struggles with multi-word slang expressions such as “no cap,” and the authors attribute this primarily to limited dataset size. We address this bot-

tleneck by constructing a larger, more balanced dataset of approximately 20,000 examples drawn from three sources. Additionally, rather than standard BERT, we fine-tune BERTweet (Nguyen et al., 2020), which is pre-trained on 850M tweets, which we hypothesize is better suited to the informal register of social media slang.

3 Problem Description

We formulate slang detection as two complementary tasks over social media text. The first is **sentence-level binary classification**, where given an input sentence x , the model predicts a label $y \in \{\text{SLANG}, \text{NOT_SLANG}\}$ indicating whether the sentence contains at least one slang expression. The second is **token-level sequence labeling**, where given a tokenized sentence $x = (x_1, x_2, \dots, x_n)$, the model predicts a corresponding BIO tag sequence $y = (y_1, y_2, \dots, y_n)$ where each $y_i \in \{\text{B-SLANG}, \text{I-SLANG}, \text{O}\}$, identifying the precise span of each slang expression.

Both tasks are complicated by two key linguistic phenomena. First, slang exhibits polysemy, where a term may carry a slang meaning in one context and a literal meaning in another (e.g., “fire” as praise vs. combustion, “ghost” as a verb meaning to cut off contact vs. a supernatural entity). Models must therefore rely on contextual understanding rather than simple lexical lookup. Second, slang vocabulary evolves rapidly, with new terms emerging and existing terms shifting in meaning across time and platforms. This limits the effectiveness of static lexicons and motivates the use of contextually-aware models.

To evaluate how different modeling paradigms handle these challenges, we implement and compare four approaches: a rule-based dictionary lookup baseline, a classical TF-IDF with logistic regression classifier, a fine-tuned BERTweet transformer model, and large language model prompting using few-shot and chain-of-thought strategies.

4 Methodology

4.1 Datasets

GenZ Slang Dataset. The GenZ Slang dataset (MLBtrio, 2024) contains approximately 1,700 entries consisting of a slang term, its definition, and an example sentence demonstrating its usage. The dataset provides curated examples that clearly

illustrate the meaning of many Gen Z slang expressions, which helps establish semantic understanding of slang terms. However, the dataset typically provides only a single example for each term, limiting the diversity of contextual usage available for training models that rely on large amounts of text.

Generation Alpha Slang Tweets Dataset. This dataset contains approximately 25,000 tweets featuring contemporary slang expressions, providing naturalistic conversational text in which slang tokens appear in authentic syntactic contexts. However, the dataset does not explicitly label whether terms are used in a slang or literal sense. To address this, we developed an automatic annotation pipeline using the Gemini 3.1 Flash API to assign BIO tags in batches of 50, balancing labeling speed with accuracy. The resulting annotations were post-processed to resolve sentence-tag length mismatches and standardize the BIO format for downstream use. Although this dataset targets Generation Alpha slang, it is relevant to our task due to substantial lexical overlap between Gen Alpha and Gen Z vernacular, as slang terms frequently circulate across adjacent age cohorts through social media platforms. However, this overlap is imperfect, as not all Gen Alpha slang is characteristic of Gen Z usage.

Wall Street Journal (WSJ) Dataset. As a source of non-slang examples, we use Wall Street Journal sentences drawn from the Penn Treebank (Marcus et al., 1999), obtained via the preprocessed release of Mohammed and Shafqat (2024). They provide sentence-level labels with all entries classified as non-slang. We performed additional pre-processing and added token-level BIO sequence labels, with all tokens tagged as O, to align the data with our task format. The WSJ sentences serve as a formal-register negative class that complements the informal negative examples from the Generation Alpha dataset, ensuring the model is exposed to diverse non-slang text.

4.2 Dataset Construction

To construct the final dataset, we combined examples from all three sources with the goal of achieving balanced class representation. From the GenZ Slang dataset, we included 1,500 usable examples as positive instances. From the Generation Alpha dataset, we selected a subset of positive examples and retained all negative examples. From the WSJ Penn Treebank, we randomly sampled

sentences as additional negative instances to ensure the model encounters formal-register text as a complement to the informal negatives from Generation Alpha.

The dataset was divided into train, development, and test splits at a 70/15/15 ratio, stratified by both label and source to ensure consistent distributions across all splits. Table 1 summarizes the resulting statistics.

	Train	Dev	Test	Gen. Test
<i>Label Distribution</i>				
SLANG	7,000	1,500	1,500	1,000
NOT_SLANG	7,000	1,500	1,500	—
<i>Source Distribution</i>				
Gen Alpha	10,993	2,355	2,356	1,000
Gen Z	1,050	225	225	—
WSJ	1,957	420	419	—
Total	14,000	3,000	3,000	1,000

Table 1: Dataset splits. Primary splits are balanced (50/50); the generalization set contains only unseen Gen Alpha slang.

We verified that no examples overlap between any splits. We additionally held out a separate generalization test set of 1,000 positive examples from the Generation Alpha dataset, excluded from the main dataset entirely. This set allows us to evaluate model performance on slang expressions not seen during training.

5 Modeling Approaches

We formulate Gen Z slang detection under two task settings: (1) sentence-level binary classification, which predicts whether a sentence contains slang, and (2) token-level sequence labeling, which identifies the full span of slang expressions using BIO tags.

To systematically evaluate modeling strategies, we implement and compare four approaches: a rule-based method, a classical machine learning model, a supervised neural model, and LLM prompting.

5.1 Rule-Based Baseline (Dictionary Lookup)

We implement a lexicon-based classifier using slang terms extracted from the GenZ Slang dataset. We construct a slang vocabulary by collecting all entries in the `Slang` column with normalization (lowercasing, trimming whitespace, and collapsing spaces), deduplicated, and sorted in descending order of length to prioritize multi-word expressions.

To prevent overlapping matches, we adopt a longest-match-first strategy, where labeled spans cannot be overwritten by shorter matches. At inference, the classifier performs exact string matches; matched spans receive B-SLANG/I-SLANG tags and the sentence is labeled SLANG. This approach is training-free and fully interpretable, but cannot handle novel slang, polysemy, or spelling variants outside the lexicon.

5.2 Classical ML Baseline (TF-IDF + Logistic Regression)

As a non-neural baseline, we implement a sentence-level classifier using TF-IDF features with logistic regression, treating each sentence as a bag of words. The motivation is to assess how much predictive signal can be extracted from surface-level lexical distributions alone, without any contextual modeling. Slang terms tend to be statistically distinctive from formal-register vocabulary, so TF-IDF weighting naturally assigns high scores to rare, slang-specific tokens while down-weighting common words shared across both classes.

We use a scikit-learn pipeline with:

- **TF-IDF:** lowercase text, unigram and bigram features (`ngram_range=(1, 2)`), `min_df=2`, and sublinear term-frequency scaling.
- **Classifier:** logistic regression with L2 regularization, `lbfgs` solver, 2,000 iterations, and a fixed random seed.

The vocabulary and IDF weights are learned on the training split and reused for development and test data. We also save predictions, gold labels, and exact-match indicators, and inspect the highest- and lowest-weighted features for analysis.

This approach serves as a strong classical baseline but does not capture long-range dependencies or contextual semantics. Because the model operates on sentence-level TF-IDF features in a bag-of-words representation, it cannot perform token-level sequence labeling.

5.3 Supervised Neural Model (BERTweet)

For contextual modeling, BERTweet (transformer-based model pretrained on large-scale Twitter data) is fine-tuned to adapt the model for sequence

labeling. Unlike bag-of-words models, this approach leverages attention-based contextual representations to capture semantic differences and disambiguate slang usage in surrounding context. This model serves as the primary supervised neural baseline for both sentence-level and token-level detection.

Sentence-Level Classification. We fine-tune BERTweet for sentence-level binary slang classification, framing the task as a two-way `SLANG` versus `NOT_SLANG` prediction problem. The implementation uses the Hugging Face `Trainer` pipeline with BERTweet’s tweet-normalized tokenizer and a sequence-classification head initialized on top of the pretrained encoder. Input tweets are tokenized with fixed-length padding and truncation, and model selection is performed on the development set using accuracy, precision, recall, and F1.

We conduct a grid search over learning rate, batch size, maximum sequence length, and number of epochs, then retrain the model using the best configuration for final evaluation on both the in-domain test set and a held-out generalization set. The best hyperparameters are a learning rate of 2×10^{-5} , batch size 16, maximum sequence length 64, and 3 training epochs.

This pipeline is designed to exploit BERTweet’s pretraining on social-media text while using dev-set F1 for model selection to better reflect the class-sensitive nature of slang detection. The relatively short sequence length and moderate training budget are also well matched to the brevity of tweet-like inputs, improving efficiency without sacrificing performance.

Token-Level Sequence Labeling. We fine-tune BERTweet for token-level BIO tagging by adding a linear classification head over its final hidden states. Because BERTweet uses a byte-pair encoding tokenizer that may split words into multiple subword pieces, we assign each word’s label to its *first* subword token only; subsequent subword tokens are masked with `IGNORE_INDEX` (-100) and do not contribute to the loss. Sentence-level predictions are derived by checking whether any token in the sequence is predicted as slang.

Training uses the Hugging Face `Trainer` API with the following hyperparameters: learning rate 2×10^{-5} , batch size 16, weight decay 0.01, maximum sequence length 128, and 3 training epochs. Model checkpoints are evaluated after each epoch; the best checkpoint is selected using token-level

F1 over slang labels (`B-SLANG`, `I-SLANG`) on the development set.

5.4 LLM-Based Methods (Few-Shot and Chain-of-Thought Prompting)

Two large language models, Qwen 2.5 Plus and Deepseek V3.2, are evaluated in a prompting-based framework without task-specific fine tuning. In the few-shot setting, the models are provided with a small number of in-context examples. Through chain-of-thought prompting, the models are encouraged to elicit step-by-step slang reasoning prior to making predictions. This approach allows comparison of prompt-based reasoning against supervised learning methods at both sentence-level and token-level classification.

Both models are accessed through Alibaba Cloud’s DashScope OpenAI-compatible API endpoint. All inference uses greedy decoding (temperature = 0) with a maximum of 512 output tokens and a fixed random seed of 42. Requests are rate-limited with a 0.2-second inter-call delay and up to 3 retries with exponential backoff on failure.

We implement four prompt template variants for each model:

Sentence-Level Templates. The system message defines the task, specifies the label vocabulary (`SLANG / NOT_SLANG`), and requires the model to output a `Label:` line. The *few-shot* variant provides 5 hand-crafted in-context examples covering unambiguous slang, polysemous words used literally, multi-word slang, and acronyms. The *chain-of-thought* (CoT) variant prepends the instruction “First, reason step-by-step about each token before assigning a label,” generating a `Reasoning:` block before the final label.

Token-Level Templates. The system message additionally specifies BIO output format and requires a `Tags:` line containing exactly n space-separated tags where n is the number of whitespace-delimited tokens. A token count hint is appended to each query. The same five in-context examples are augmented with BIO tag sequences, and the CoT variant again elicits step-by-step reasoning before the tag sequence.

Responses are parsed with a rule-based extractor that locates the `Label:`, `Tags:`, and `Reasoning:` fields via regex. Tag sequences with length mismatches are corrected by truncation or padding with `O` tags; the sentence-level label is derived from the tag sequence when a token-level template is used.

6 Results

We evaluate all methods under two task settings: (1) sentence-level binary classification and (2) token-level sequence labeling using BIO tags. Performance is measured using accuracy, precision, recall, and F1-score where appropriate.

6.1 Evaluation Setup

We evaluate model performance on a held-out test set consisting of 3,000 examples balanced evenly between SLANG and NOT_SLANG labels. This balanced distribution enables reliable comparison across accuracy, precision, recall, and F1-score.

To assess robustness to unseen slang, we additionally evaluate models on a generalization test set containing 1,000 examples composed entirely of novel slang expressions. This dataset is intentionally imbalanced and is used to measure recall-oriented generalization performance rather than overall classification accuracy.

6.2 Sentence-Level Classification Results

Table 2 summarizes the performance of all models on the sentence-level slang detection task.

Model	Accuracy	Precision	Recall	F1
Dictionary Lookup	0.600	0.567	0.840	0.677
TF-IDF + Logistic Regression	0.847	0.847	0.846	0.847
Fine-tuned BERTweet	0.908	0.882	0.942	0.911
DeepSeek V3.2 (CoT)	0.801	0.740	0.928	0.823
DeepSeek V3.2 (Few-shot)	0.788	0.720	0.944	0.817
Qwen 2.5 Plus (CoT)	0.815	0.781	0.874	0.825
Qwen 2.5 Plus (Few-shot)	0.793	0.733	0.923	0.817

Table 2: Sentence-level slang detection performance.

Overall, neural and LLM-based approaches outperform rule-based and classical baselines. The dictionary lookup achieves high precision but suffers from low recall due to limited coverage of emerging slang. TF-IDF improves recall but struggles with context-dependent expressions. BERTweet achieves the best balance across metrics, benefiting from contextual representations. Prompted LLMs show competitive performance, particularly in recall, though results vary depending on prompting strategy.

6.3 Token-Level Sequence Labeling Results

Table 3 reports performance on the token-level BIO tagging task.

The fine-tuned BERTweet model achieves the strongest token-level performance, demonstrating the advantage of supervised sequence labeling. Dictionary lookup remains highly precise but fails

Model	Accuracy	Precision	Recall	F1
Dictionary Lookup	0.923	0.273	0.521	0.358
TF-IDF + Logistic Regression	—	—	—	—
Fine-tuned BERTweet	0.983	0.810	0.761	0.785
DeepSeek V3.2 (CoT)	0.936	0.343	0.652	0.449
DeepSeek V3.2 (Few-shot)	0.932	0.280	0.476	0.352
Qwen 2.5 Plus (CoT)	0.953	0.438	0.569	0.495
Qwen 2.5 Plus (Few-shot)	0.927	0.253	0.442	0.322

Table 3: Token-level slang span detection (BIO tagging).

to capture multi-token spans and novel slang. LLM-based approaches show moderate performance, often identifying the presence of slang but struggling with precise span boundaries.

6.4 Comparison Across Approaches

Across both tasks, clear differences emerge between modeling paradigms. Rule-based methods provide strong precision but lack generalization. Classical machine learning models capture surface-level lexical patterns but fail on contextual ambiguity. Transformer-based models offer the best overall performance due to their ability to encode context and handle sequence structure. Prompt-based LLMs demonstrate strong zero/few-shot capabilities, particularly for sentence-level detection, but are less reliable for fine-grained token classification.

6.5 Generalization to Novel Slang

Model	Recall	F1
Dictionary Lookup	0.768	0.869
TF-IDF + Logistic Regression	0.787	0.881
Fine-tuned BERTweet	0.768	0.869
DeepSeek V3.2 (CoT)	0.936	0.967
DeepSeek V3.2 (Few-shot)	0.946	0.972
Qwen 2.5 Plus (CoT)	0.875	0.933
Qwen 2.5 Plus (Few-shot)	0.906	0.951

Table 4: Generalization test set performance (1,000 novel slang sentences; precision = 1.0 for all models).

Table 4 reports performance on the generalization test set, which contains 1,000 novel slang examples not present in training. Because all examples are positive, precision is 1.0 for all models and performance is governed entirely by recall.

Notably, LLMs substantially outperform supervised models on this set, reversing the ranking seen on the balanced test set. DeepSeek V3.2 achieves the highest recall (0.946 few-shot), followed by Qwen 2.5 Plus (0.906 few-shot). Fine-tuned BERTweet, despite its strong in-domain per-

formance, achieves only 0.768 recall, on par with the dictionary baseline. This suggests that supervised fine-tuning leads to some degree of memorization of training-set slang patterns that does not transfer to entirely unseen vocabulary. In contrast, LLMs leverage broad pretraining knowledge of informal internet language, enabling them to recognize novel slang terms not seen during any task-specific training. Interestingly, few-shot prompting outperforms CoT on the generalization set for both models, likely because CoT reasoning sometimes over-rationalizes ambiguous expressions as standard English, suppressing recall. The token-level generalization results follow a similar pattern: BERTweet token F1 drops substantially to 0.609, while Qwen and DeepSeek CoT achieve token F1 of 0.391 and 0.385 respectively, indicating that precise span boundary identification remains difficult for all systems on out-of-distribution slang.

6.6 Error Analysis

Error analysis reveals that the primary challenge is not identifying slang vocabulary, but determining whether a potentially slang-bearing form is being used as slang in context. As shown in Table 5 in Appendix A, performance improves from lexical matching to contextual modeling, yet every system remains sensitive to ambiguity, novelty, and annotation assumptions.

The dictionary baseline relies on surface-form heuristics, leading to frequent over-prediction on ambiguous words and missed multiword or emerging slang expressions, especially in the BIO setting. The TF-IDF classifier performs substantially better, indicating that distributional lexical cues capture much of the task, but it still struggles with sparse, rare, or domain-shifted slang.

BERT-based models are the most robust overall. The sentence-level BERTweet classifier achieves the best precision-recall balance, reflecting the benefits of contextual pretraining on social-media text for disambiguation between literal and slang senses. However, it still produces false positives on literal uses and false negatives on short, low-context, or highly novel inputs. At the token level, BERTweet improves over the dictionary BIO baseline, but exact span detection remains challenging due to partial matches, span fragmentation, and boundary drift in multi-token slang expressions.

Polysemy analysis in Table 6 in Appendix A highlights consistently difficult terms such as

fire, big, beta, and ghost, while basic, extra, woke, squad, mood, and lore also remain difficult in most settings. Error types vary by model: lexicon-driven systems over-trigger on frequent ambiguous items such as basic, extra, and cringe; TF-IDF often misses non-canonical or weakly lexicalized slang uses; and LLM-based sentence classifiers tend to rationalize ambiguous slang as literal or domain-specific language, leading to conservative predictions and recall loss.

Overall, findings suggest that future gains will depend less on expanding slang vocabularies, but on improving contextual sense disambiguation, multiword span modeling, and robustness to evolving usage.

7 Conclusion and Future Work

Using a balanced dataset of approx. 20,000 examples from three sources, we evaluated a dictionary lookup baseline, TF-IDF with logistic regression, fine-tuned BERTweet, and prompting-based LLMs (DeepSeek V3.2 and Qwen 2.5 Plus). Fine-tuned BERTweet achieved the best in-domain performance across both tasks (sentence-level F1: 0.911; token-level F1: 0.785), while LLMs demonstrated advantage on the generalization test set, with DeepSeek V3.2 reaching a recall of 0.946 on unseen slang. Error analysis revealed that polysemy remains the primary challenge across all systems, with words such as fire, big, and basic consistently causing errors, highlighting the difficulty of context-sensitive disambiguation in informal language.

Future work includes extending the binary labeling scheme to multiclass classification (e.g., distinguishing between generational origins of slang such as Gen Z vs. Gen Alpha) and incorporating multimodal signals such as emojis and hashtags to capture richer context and finer-grained distinctions in informal language evolution.

Finally, the complementary strengths of supervised models and LLMs motivate hybrid approaches: BERTweet excels on in-distribution examples, while LLMs generalize better to novel vocabulary. Exploring hybrid approaches that combine the contextual precision of fine-tuned transformers with the broad pretraining knowledge of LLMs — for instance, through LLM-assisted data augmentation or retrieval-augmented prompting — offers a promising path toward more robust slang detection.

8 Individual Contributions

Christopher Ernesto: Acquired and preprocessed the Gen Alpha slang tweets dataset. Implemented a BIO-tagging pipeline to automate dataset annotation using Gemini 3.1 Flash. Collaborated with Christopher Sumali to fine-tune BERTweet for token-level BIO tagging by implementing the data preparation pipeline, transforming the annotated dataset into a format compatible with token-level classification tasks. Performed error analysis on token-level predictions for BERTweet to evaluate model performance with a focus on polysemous words.

Ryan Keng: Led project management efforts and facilitated sprint planning sessions to delegate work across the team. Wrote processing scripts to generate sentence-level binary labels for Gen Z and Gen Alpha datasets. Built the unified dataset pipeline, combining sources and performing a stratified 70/15/15 train/dev/test split. Fine-tuned BERTweet for sentence-level binary slang classification. Performed error analysis on sentence-level predictions from dictionary and TF-IDF baselines.

Jasmine Lao: Implemented a dictionary-based lookup baseline for an initial comparison point, and produced evaluation reports for this baseline. Conducted error analysis on sentence-level BERTweet model, identifying common failure cases and systematic patterns. Generated evaluation reports and produced prediction outputs for sentence-level BERTweet model to support comparative analysis. Contributed to codebase maintenance through reviewing and approving pull requests.

Romeo Nickel: Acquired and preprocessed the GenZ Slang (MLBTrio) dataset, implementing a BIO tagging pipeline with text normalization. Designed and implemented the LLM evaluation pipeline, including four prompt template variants (sentence and token-level few-shot and chain-of-thought), an API client for Qwen 2.5 Plus and DeepSeek V3.2 via DashScope, robust response parsing, and metric computation. Evaluated both models across all four templates on the test and generalization sets. Performed qualitative analysis of chain-of-thought rationales and conducted polysemy-focused error analysis for LLM models on both test and generalization sets.

Christopher Sumali: Implemented the TF-IDF + logistic regression sentence-level classifier, including the scikit-learn pipeline, preprocessing de-

sign, hyperparameter selection, and feature importance analysis. Fine-tuned BERTweet for token-level BIO tagging, including subword label alignment, training configuration, checkpoint selection on the development set, and evaluation on both the standard and generalization test sets. Conducted token-level error analysis for the dictionary lookup baseline, identifying key failure modes such as polysemy, span boundary errors, and novel slang misses.

9 Code Availability

The code and trained models used in this work are publicly available at <https://github.com/jsmnlao/GenZ-Slang-Detection>.

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Appendix

A Error Analysis

Table 5: Summary of polysemous-word behavior across reports. “Correct” counts words marked as error-free in a report’s polysemy section; “Errors” counts words with at least one false positive or false negative. “FP-heavy” and “FN-heavy” list the most salient over-triggered and missed polysemous items in each report.

Report	Correct	Errors	Main FP-heavy words	Main FN-heavy words
<i>Test Set</i>				
Dictionary Sentence (Test)	0	15	basic, extra, squad	beta, fire, big
Dictionary BIO (Test)	3	12	basic, extra, woke	big, beta, fire
TF-IDF Sentence (Test)	0	15	big, cringe, slay	beta, basic, woke
BERT Sentence (Test)	3	12	big, fire, squad	basic, mood, lore
BERT BIO (Test)	3	12	big, cringe, slay	extra, cringe, big
Qwen (Test)	0	15	lore, squad, mood	woke, big, basic
DeepSeek (Test)	0	15	lore, squad, beta	extra, big, basic
<i>Generalization Test Set</i>				
Dictionary Sentence (Gen.)	6	8	–	beta, big, mood
Dictionary BIO (Gen.)	2	12	cringe, sus, woke	big, fire, mood
TF-IDF Sentence (Gen.)	0	14	–	beta, big, fire
BERT Sentence (Gen.)	1	13	–	beta, fire, extra
BERT BIO (Gen.)	3	11	cringe, sus, big	fire, big, beta
Qwen (Gen.)	5	10	–	woke, big, beta
DeepSeek (Gen.)	9	6	–	big, fire, beta

Table 6: Cross-report difficulty of polysemous words. “Error reports” indicates in how many of the 14 report sections a word produced at least one error; “FP reports” and “FN reports” count the number of report sections in which the word contributed false positives or false negatives, respectively.

Word	Error reports	FP reports	FN reports	Aggregate FP	Aggregate FN
fire	14/14	7	14	64	49
big	14/14	7	14	76	129
beta	14/14	6	14	76	59
ghost	14/14	6	14	35	27
basic	13/14	9	9	192	29
mood	12/14	5	10	64	30
squad	12/14	4	10	86	17
woke	12/14	8	6	55	27
cringe	12/14	9	7	65	14
extra	11/14	7	8	192	25
sus	11/14	9	4	39	9
slay	10/14	8	3	29	5
lore	9/14	4	8	91	17
sick	8/14	3	6	5	10